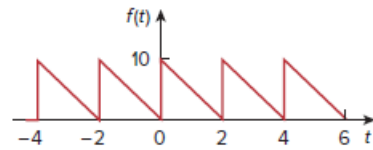


Problem

Find the Fourier series expansion of the backward sawtooth waveform of Fig. 17.48. Obtain the amplitude and phase spectra.

Figure 17.48:



Step-by-step solution

Step 1 of 8

Refer to Figure 17.48 in the textbook.

The expression for the sawtooth backward waveform for one period is,

$$f(t) = 5(2 - t) \quad 0 \leq t \leq 2$$

The time period of the signal is,

$$T = 2$$

Calculate the value of ω_0 .

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ &= \pi \text{ rad/s} \end{aligned}$$

Step 2 of 8

Calculate the Fourier coefficient, a_0 .

$$\begin{aligned} a_0 &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{2} \int_0^2 5(2 - t) dt \\ &= \frac{5}{2} \left[2t - \frac{t^2}{2} \right]_0^2 \\ &= \frac{5}{2} \left[2(2) - \frac{2^2}{2} - 0 \right] \\ a_0 &= 5 \end{aligned}$$

Step 3 of 8

Calculate the Fourier coefficient, a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{2} \int_0^2 5(2-t) \cos(n\pi t) dt \\
 &= \int_0^2 [10 \cos(n\pi t) - 5t \cos(n\pi t)] dt \\
 &= 10 \int_0^2 \cos(n\pi t) dt - 5 \int_0^2 t \cos(n\pi t) dt \\
 &= 10 \left[\frac{\sin n\pi t}{n\pi} \right]_0^2 - 5 \left[\frac{n\pi t \sin(n\pi t) + \cos(n\pi t)}{(n\pi)^2} \right]_0^2 \\
 &= \frac{10}{n\pi} [\sin 2n\pi - \sin 0] - \frac{5}{(n\pi)^2} [n\pi(2) \sin(2n\pi) + \cos(2n\pi) - 0 - \cos(0)] \\
 &= \frac{10}{n\pi} [0 - 0] - \frac{5}{(n\pi)^2} [0 + 1 - 0 - 1] \\
 &= 0
 \end{aligned}$$

Step 4 of 8

Calculate the Fourier coefficient, b_n .

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt \\
 &= \frac{2}{2} \int_0^2 5(2-t) \sin(n\pi t) dt \\
 &= \int_0^2 [10 \sin(n\pi t) - 5t \sin(n\pi t)] dt \\
 &= 10 \int_0^2 \sin(n\pi t) dt - 5 \int_0^2 t \sin(n\pi t) dt \\
 &= 10 \left[-\frac{\cos n\pi t}{n\pi} \right]_0^2 - 5 \left[\frac{\sin(n\pi t) - n\pi t \cos(n\pi t)}{(n\pi)^2} \right]_0^2 \\
 &= -\frac{10}{n\pi} [\cos 2n\pi - \cos 0] - \frac{5}{(n\pi)^2} [\sin(2n\pi) - 2n\pi \cos(2n\pi) - \sin(0) + 0] \\
 &= -\frac{10}{n\pi} [1 - 1] - \frac{5}{(n\pi)^2} [0 - 2n\pi - 0 - 1] \\
 &= \frac{10}{n\pi}
 \end{aligned}$$

[Comments \(1\)](#)



Anonymous

2nd to last part should be [0-2npi-0-0]

Step 5 of 8

Calculate the expression for amplitude spectra.

$$\begin{aligned}
 A_n &= \sqrt{a_n^2 + b_n^2} \\
 &= \sqrt{(0)^2 + \left(\frac{10}{n\pi}\right)^2} \\
 &= \frac{10}{n\pi}
 \end{aligned}$$

Step 6 of 8

Calculate the expression for phase spectra.

$$\begin{aligned}\phi_n &= -\tan^{-1}\left(\frac{b_n}{a_n}\right) \\ &= -\tan^{-1}\left(\frac{\frac{10}{n\pi}}{0}\right) \\ &= -90^\circ\end{aligned}$$

Calculate the Fourier series expansion.

$$\begin{aligned}f(t) &= a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n) \\ &= 5 + \sum_{n=1}^{\infty} \left(\frac{10}{n\pi}\right) \cos(n\omega_0 t - 90^\circ)\end{aligned}$$

Thus, the Fourier series expansion of the waveform is $5 + \sum_{n=1}^{\infty} \left(\frac{10}{n\pi}\right) \cos(n\omega_0 t - 90^\circ)$.

Step 7 of 8

Draw the amplitude spectrum of the waveform.

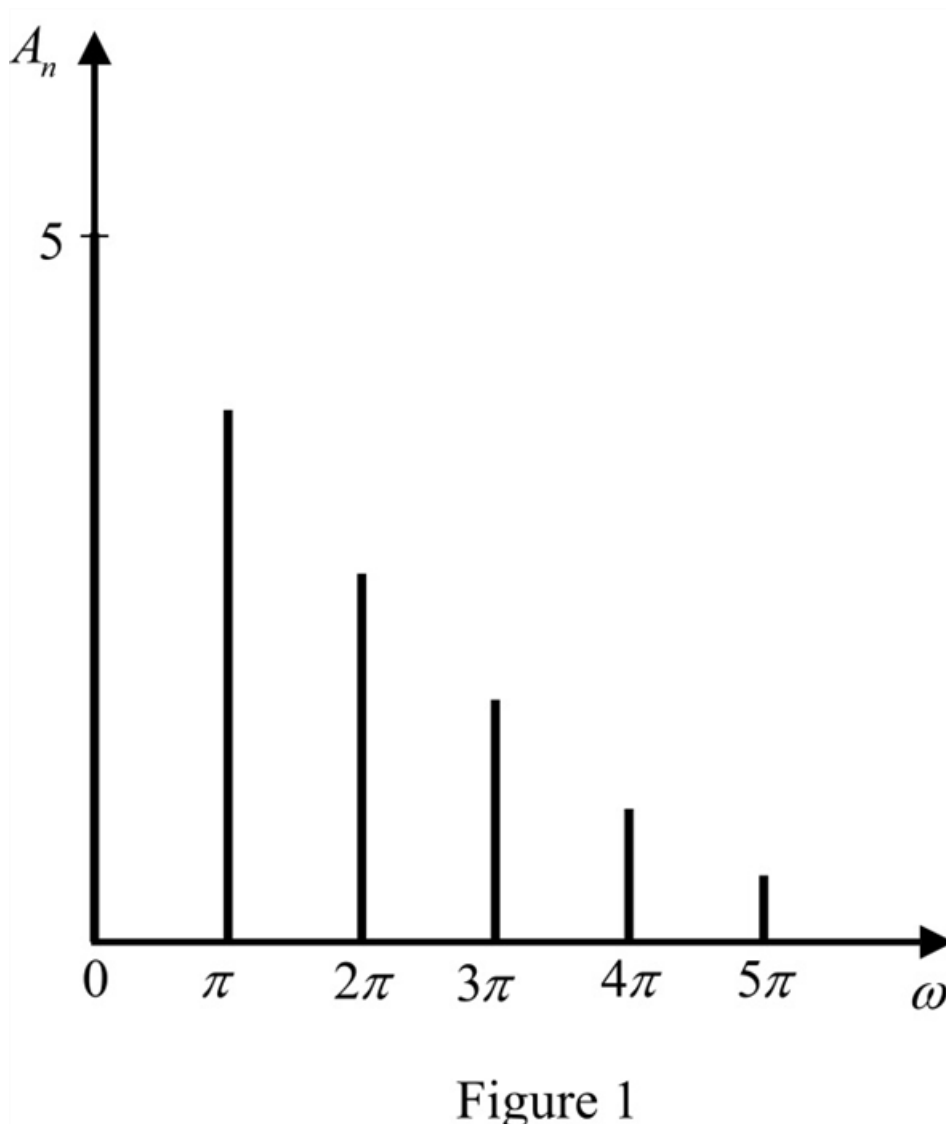


Figure 1

Step 8 of 8

Draw the phase spectrum of the waveform.

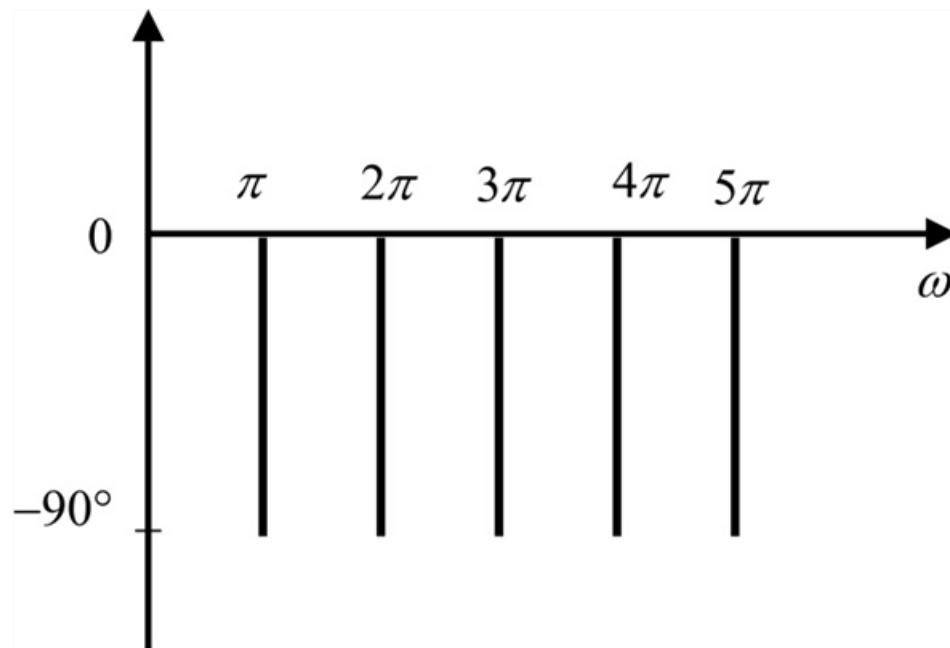


Figure 2

Thus, the amplitude and phase spectrum of waveforms is shown Figures 1 and 2.